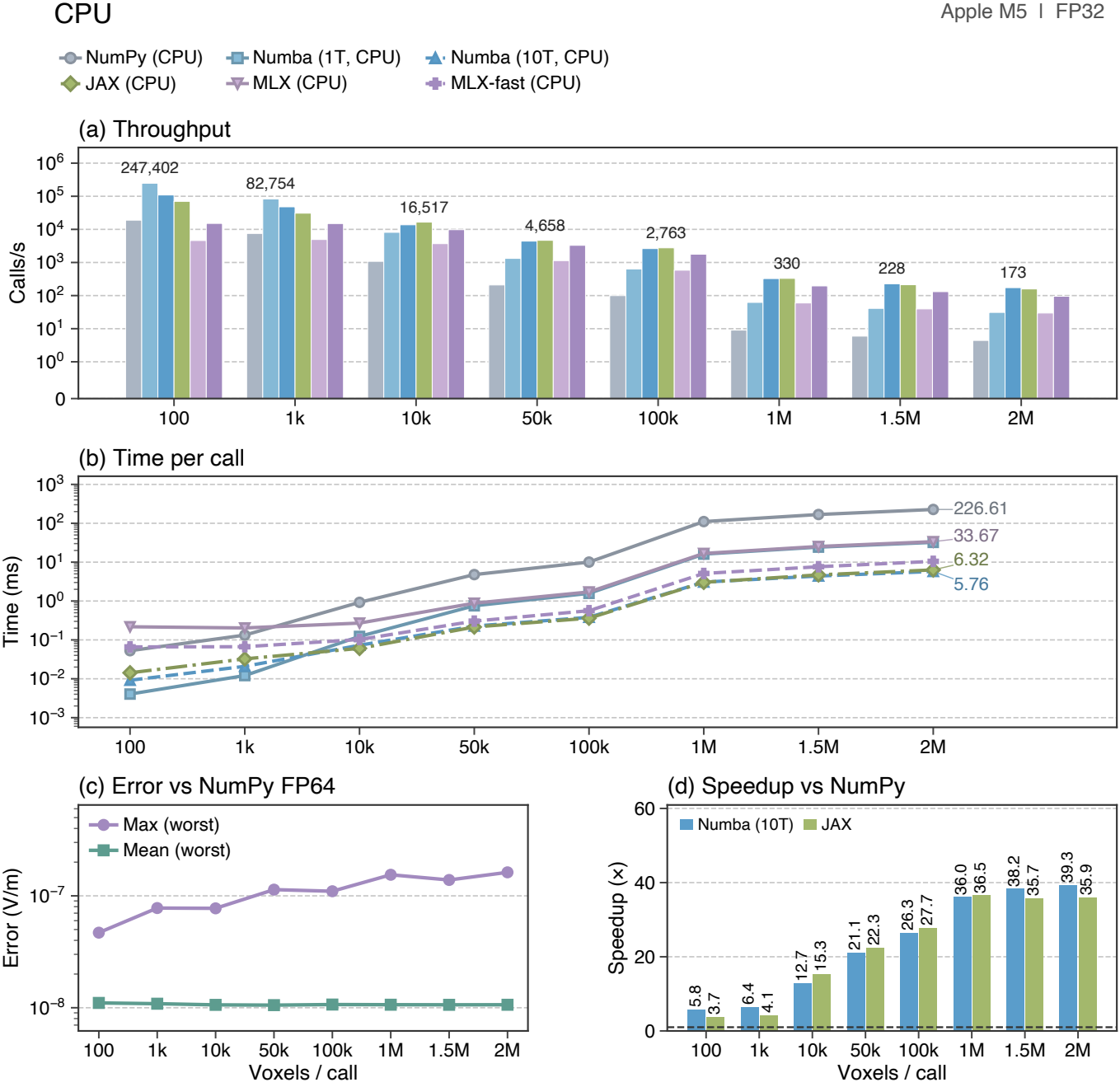
NumPy (CPU)
Numba (1T, CPU)
Numba (10T, CPU)
JAX (CPU)
MLX (CPU)
MLX-fast (CPU)
(a) Throughput
Calls/s
247,402
82,754
16,517
4,658
2,763
330
228
173
100
1k
10k
50k
100k
1M
1.5M
2M
(b) Time per call
Time (ms)
226.61
33.67
6.32
5.76
(c) Error vs NumPy FP64
Max (worst)
Mean (worst)
Error (V/m)
Voxels / call
(d) Speedup vs NumPy
Numba (10T)
JAX
Speedup (×)
5.8
3.7
6.4
4.1
12.7
15.3
21.1
22.3
26.3
27.7
36.0
36.5
38.2
35.7
39.3
35.9
Voxels / call